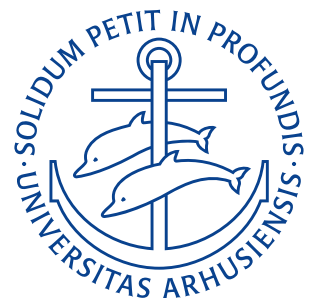


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Assignment 1: State of Stress

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Definition of The Problem

Let us consider a continuum and a Cartesian reference system $(O; \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$. In this system, the Cauchy stress tensor has the following representation:

$$[\boldsymbol{\sigma}] = \begin{pmatrix} x_1^2 + x_2^2 & x_2^2 & x_3^2 + x_1 x_2 \\ x_2^2 & 0 & 2x_3^2 \\ x_3^2 + x_1 x_2 & 2x_3^2 & x_3^2 - x_1^2 \end{pmatrix},$$

where (x_1, x_2, x_3) are the Cartesian coordinates of a point within the continuum. So for example, the position vector of a point $\mathbf{P}$ can be written as $\mathbf{P} = x_i \mathbf{e}_i$. Its components can be arranged in an array $\mathbf{x}$.

Problem 1

Consider three material points on the continuum

- $\mathbf{P}_1$, with coordinates: $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$;
- $\mathbf{P}_2$, with coordinates: $\begin{pmatrix} 1 & 1 & 2 \end{pmatrix}$;
- $\mathbf{P}_3$, with coordinates: $\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$;

and perform the following for each of the material points above:

- compute the principal stresses and the principal directions of stress
- form the rotation matrix that diagonalizes the Cauchy stress tensor, once represented in the bases of the eigenvectors
- describe the state of stress: Is it plane stress? Hydrostatic stress?

Derivation

We start by computing the three stress tensors

$$\boldsymbol{\sigma}_1 = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 8 \\ 5 & 8 & 3 \end{pmatrix} \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

$\mathbf{P}_1$ To find the principal stresses we must solve the eigenvalue problem:

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0.$$

Which for $\boldsymbol{\sigma}_1$ is:

$$\begin{vmatrix} 2-\lambda & 1 & 2 \\ 1 & 0-\lambda & 2 \\ 2 & 2 & 0-\lambda \end{vmatrix} = 0$$

$$-1\lambda^3 + 2\lambda^2 + 9\lambda = 0.$$

Solving this gives $\lambda_1 = 0, \lambda_2 = 1 - \sqrt{10}, \lambda_3 = 1 + \sqrt{10}$.

To find the principal directions we solve the eigenvector problem:

$$(\boldsymbol{\sigma} - \lambda \mathbf{I})\mathbf{v}^{(j)} = 0.$$

For $\lambda_1 = 0$ we get:

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \implies \mathbf{v}^{(1)} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}.$$

Which normalizes to:

$$\hat{\mathbf{v}}^{(1)} = \frac{\mathbf{v}^{(1)}}{|\mathbf{v}^{(1)}|} = \frac{\begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}}{\sqrt{2^2 + 2^2 + 1^2}} = \begin{pmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}.$$

For $\lambda_2 = 1 - \sqrt{10}$ we get

$$\begin{pmatrix} 2 - 1 - \sqrt{10} & 1 & 2 \\ 1 & 0 - 1 - \sqrt{10} & 2 \\ 2 & 2 & 0 - 1 - \sqrt{10} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \implies \mathbf{v}^{(2)} = .$$

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} \rightarrow \left\{ \frac{1}{4}(2 + \sqrt{10}), \frac{\sqrt{\frac{5}{2}}}{2}, 1 \right\} \quad \left\{ \frac{1}{4}(2 - \sqrt{10}), -\frac{\sqrt{\frac{5}{2}}}{2}, 1 \right\} \quad \{-2, 2, 1\} .$$

Problem 2

Compute the traction vector at the point $\mathbf{P}_i$ on the element of surface with unit normal $\mathbf{n}_i$ for the following cases:

- $\mathbf{P}_1$ and $\mathbf{n}_1 = \begin{pmatrix} -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{pmatrix}$;
- $\mathbf{P}_2$ and $\mathbf{n}_2 = \begin{pmatrix} 1 & 1 & 0 \end{pmatrix} / \sqrt{2}$;
- $\mathbf{P}_3$ and $\mathbf{n}_3 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} / \sqrt{3}$.

Problem 3

Determine whether the stress field above satisfies the equilibrium (balance) equations in the absence of body forces. Otherwise, compute the body forces that the continuum needs to be subjected to, so as to be in equilibrium.